

Smart Wagon — Gateway Telemetry Protocol

MQTT Topics & JSON Message Specification

Gateway / Data Concentrator (Class A) • Uplink + Downlink
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Document status — living specification

This is a working specification under active development. Message types, field names, enumerations, topic structure and timing values are expected to change by up to ~30% as development progresses. Changes will be reflected in updated revisions of this document; treat any external integration built against it as subject to revision until the protocol is marked final.

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1. Overview & Conventions

This document defines every message exchanged between the Smart Wagon gateway (data concentrator) and the cloud platform, the JSON payload of each, and the MQTT topic tree carrying them. It is the integration contract between the device and the cloud/application layer. Thresholds, alarm logic and band calibration are governed by RDSO **WD-35-MISC-2024** §7.15 and are configurable at runtime via downlink command.

1.1 Message classes

Four message types travel uplink (gateway → cloud) and one travels downlink (cloud → gateway):

mt	Type	Direction	Trigger / cadence	Priority
alarm	Alarm	Uplink	Safety event — immediate, breaks the cycle	Highest
event	Event	Uplink	Discrete state change (train start/stop, etc.)	High
hb	Heartbeat	Uplink	Periodic: 10 min running / 12 h stopped	Normal
resp	Response	Uplink	Reply to a queued downlink command	High
cmd	Command	Downlink	Cloud-issued; queued while gateway sleeps	—

1.2 Class A behaviour (why this matters for timing)

The gateway is **uplink-first, like a LoRaWAN Class A device**. It sleeps to meet the 6-year power budget and is not continuously reachable. The cloud cannot push a command at will — it **queues** the command on the broker, and the gateway drains the queue only during a wake window (after it publishes an alarm or heartbeat). Consequences:

- **Worst-case command latency = next wake:** ≤ 10 min when running, ≤ 12 h when stopped. A parked wagon does not wake early to poll for commands; queued commands are delivered on its next scheduled heartbeat.
- Critical alarms are independent of the heartbeat timer. The gateway keeps a low-power receive window open — it listens **100 ms every 5 s** (§3.6) — to catch critical alarm events, so a yard pilferage door-open (DOOR_UNAUTH) still fires within ~5 s even during the 12 h stopped gap.

1.3 Encoding conventions

Item	Convention
Serialization	UTF-8 JSON. Keys are abbreviated to conserve cellular payload; see the field dictionary per message. CBOR may be substituted later 1:1 to shrink payload further.
Timestamps	ts = UTC epoch seconds (uint32), sourced from GNSS or backend per §7.17.

Item	Convention
Location	loc = { lat, lon, cep, sys } in decimal degrees; cep = CEP50 radius in metres — true position within this radius ~50% of the time (target < 5 m, §7.17). sys = GNSS constellation(s) used for the fix (see §2.3).
Identifiers	gw = gateway ID; wgn = wagon number (the only mandatory node-provisioning key, §7.19).
Sequence	seq = monotonic uint per gateway. Buffered records are republished with their ORIGINAL ts and seq; cloud de-duplicates on (gw, seq).
Speed	spd in km/h. Units stated per field elsewhere.
Versioning	pv = protocol version (currently 1). Breaking changes bump pv and the topic version segment v1 → v2.

2. Common Envelope

Every uplink message shares one envelope; the type-specific payload lives under the `d` object.

```
{
  "mt": "alarm",           // message type: alarm | event | hb | resp
  "pv": 1,                 // protocol version
  "seq": 10432,             // monotonic sequence number (dedup key with gw)
  "gw": "GW000734",        // gateway / data concentrator ID
  "wgn": "BOXN21456789",   // wagon number (RDSO §7.19)
  "ts": 1718971200,        // UTC epoch seconds (RDSO §7.17)
  "loc": { "lat": 18.52043, "lon": 73.85674, "cep": 4, "sys": ["navic","gps"] }, // CEP50
m + GNSS source
  "spd": 0,                // ground speed km/h
  "pwr": { "src":"lto", "soc":78, "sol":"charging", "bkp":"ok" }, // compact power (§2.2)
  "d": { /* type-specific payload */ }
}
```

2.1 Envelope field dictionary

Key	Type	Description
mt	string	Message type: alarm, event, hb, resp.
pv	int	Protocol version (1).
seq	uint	Monotonic sequence number; dedup key together with gw.
gw	string	Gateway ID.
wgn	string	Wagon number (§7.19).
ts	uint	UTC epoch seconds (§7.17).
loc	obj	{ lat, lon, cep, sys } — cep = CEP50 metres (true position within radius ~50% of fixes, §7.17); sys = GNSS constellation(s) used for the fix (§2.3).
spd	num	Ground speed, km/h.
pwr	obj	Compact power summary — present on every message (see §2.2).
d	obj	Type-specific payload (§3–§6).

2.2 Compact power summary (pwr) — on every message

The power chain is **Solar (harvest)** → **LTO buffer** → **[power mux]** ← **Li-SOCl₂ cell** (backup). The system normally runs off the solar-charged LTO; the power mux falls over to the Li-SOCl₂ backup only when the LTO sags below the switchover threshold. **src** reports which side of the mux is live. A compact summary rides on every message so even a fast alarm carries the power posture; the full block is in the heartbeat (§5.3).

Key	Values	Meaning
src	lto bkp trn	Active rail: solar/LTO primary, Li-SOCl ₂ backup, or in transition (read from the power-mux status, not inferred).
soc	0–100	LTO state-of-charge %.

Key	Values	Meaning
sol	off present charging	Solar state. 'present' ≠ 'charging': in low light or cold the panel can be present but below its charging-operation threshold, so it is not actually charging.
bkp	ok low crit	Li-SOCl ₂ backup health, derived from coulomb-counted remaining capacity (NOT voltage — flat discharge curve).

2.3 GNSS source & fix provenance (sys)

The receiver is multi-constellation. Each fix records the constellation(s) that produced it in `loc.sys`, so the cloud knows the provenance of every position. The preferred constellation is **NavIC-first**; in locations where the default set cannot hold a reliable fix, the preferred set can be changed remotely with the `set_gnss` command (§6.2). Per-fix provenance is in `loc.sys`; full GNSS engine status (fix type, satellites used, HDOP) is reported in the heartbeat `gnss` block (§5.1).

sys code	Constellation	Notes
navic	NavIC / IRNSS (India)	Regional; preferred first source for Indian Railways operation.
gps	GPS (USA)	Global.
glonass	GLONASS (Russia)	Global.
galileo	Galileo (EU)	Global.
beidou	BeiDou (China)	Global.

sys is an array because a fix may combine constellations, e.g. ["navic", "gps"]. Per-message samples below may abbreviate loc to { lat, lon, cep } for brevity; sys is always permitted.

3. Alarm Message (mt = "alarm")

Safety-critical, asynchronous, breaks the reporting cycle. Published QoS 1 on the alarm topic. Each alarm maps to a configurable threshold (§7.15) and drives cloud cellular/email escalation (§7.16).

3.1 Payload (d) field dictionary

Key	Type	Description
code	string	Alarm code — see §3.2 enumeration.
sev	yellow red	Condition band for bearing/wheel (§7.10); use warn crit for non-band alarms.
node	string null	Source sensor node serial; null for gateway-level alarms.
val	num	Measured value that triggered the alarm.
unit	string	Unit of val (C, mm, bar, g, ...).
thr	num	Threshold that was breached.
st	set clear	Alarm onset (set) vs. return-to-normal (clear).
aid	string	Alarm instance ID; the matching 'clear' reuses it, and cloud ACK references it.
spd	num	Speed at trigger (km/h) — preserved in payload for buffered integrity of moving-condition alarms.

3.2 Alarm code enumeration

code	Alarm	Source	Clause
DERAIL	Derailment	IMU + load-sensor fusion	§5.3.3
HOT_BEARING	Bearing over-temp / seizure	Bearing/wheel sensor	§7.10
FLAT_WHEEL	Wheel flat / RCF defect	Bearing/wheel sensor (vibration)	§7.9, §7.13
DOOR_UNAUTH	Door open in unauthorised zone / while moving	Door sensor	§5.5.1
HANDBRAKE_MOVING	Handbrake engaged while moving	Handbrake sensor	§5.6.1
OVERLOAD	Load over configured limit	Load sensor	§5.3
TILT	Tilted / asymmetric load	Load & tilt sensor	§5.3
TANK_OVERTEMP	Tank/cargo temperature breach	Tank temp sensor	§5.4.2
IMPACT	G-shock / impact	IMU	§2.1
SENSOR_FAULT	Sensor damaged / not responding	Any node	§7.14
TAMPER	Gateway/sensor removal or tamper	Removal / tilt detection	§5.x

code	Alarm	Source	Clause
LOW_BATTERY	Gateway power critical (gateway battery only)	Power subsystem	system
NODE_LOW_BATTERY	Sensor node battery low (non-replaceable cell)	Sensor node	§7.14

Severity governs delivery. sev red / crit alarms are safety-critical and transmitted immediately on the receive window (§3.6). sev warn alarms — e.g. NODE_LOW_BATTERY — are advisory and delivered on the next scheduled heartbeat rather than waking the unit. NODE_LOW_BATTERY carries the sensor node serial in node and its remaining battery % in val; the gateway's own battery is reported separately as LOW_BATTERY with node = null.

3.3 Sample — derailment (set)

```
{
  "mt":"alarm", "pv":1, "seq":20871, "gw":"GW000734", "wgn":"BOXN21456789",
  "ts":1718971412,
  "loc":{"lat":18.51990, "lon":73.86012, "cep":4 },
  "spd":54,
  "pwr":{"src":"lto", "soc":81, "sol":"off", "bkp":"ok" },
  "d":{"
    "code":"DERAIL",
    "sev":"red",
    "node":null,
    "val":7.8, "unit":"g",
    "thr":4.0,
    "st":"set",
    "aid":"al-20871",
    "spd":54
  }
}
```

3.4 Sample — hot bearing (set, Red band)

```
{
  "mt":"alarm", "pv":1, "seq":20872, "gw":"GW000734", "wgn":"BOXN21456789",
  "ts":1718971450,
  "loc":{"lat":18.51970, "lon":73.86090, "cep":3 },
  "spd":52,
  "pwr":{"src":"lto", "soc":81, "sol":"off", "bkp":"ok" },
  "d":{"
    "code":"HOT_BEARING",
    "sev":"red",
    "node":"NODE-BRG-03",
    "val":118, "unit":"C",
    "thr":95,
    "st":"set",
    "aid":"al-20872",
    "spd":52
  }
}
```

3.5 Sample — door opened in unauthorised zone, later cleared

```
{
```



```

"mt":"alarm", "pv":1, "seq":21044, "gw":"GW000734", "wgn":"BCNA33120097",
"ts":1718990000,
"loc":{"lat":19.07601, "lon":72.87766, "cep":5 },
"spd":0,
"pwr":{"src":"bkp", "soc":12, "sol":"off", "bkp":"low" },
"d":{"code":"DOOR_UNAUTH", "sev":"crit", "node":"NODE-DOOR-01",
    "val":1, "unit":"state", "thr":0, "st":"set", "aid":"a1-21044", "spd":0 }
}

```

Sample — sensor node low battery (advisory)

```

{
  "mt":"alarm", "pv":1, "seq":21090, "gw":"GW000734", "wgn":"BOXN21456789",
  "ts":1719003000,
  "loc":{"lat":18.71010, "lon":73.99820, "cep":4, "sys":["navic","gps"] },
  "spd":47,
  "pwr":{"src":"lto", "soc":70, "sol":"charging", "bkp":"ok" },
  "d":{"code":"NODE_LOW_BATTERY", "sev":"warn", "node":"NODE-BRG-05",
    "val":12, "unit":"%", "thr":15, "st":"set", "aid":"a1-21090", "spd":47 }
}

```

3.6 Critical-alarm receive window (RX duty cycle)

To catch critical alarm events from sensor nodes without keeping the receiver powered continuously, the gateway runs a duty-cycled receive window: it **listens for 100 ms after every 5 s**. This bounds worst-case alarm-detection latency to about 5 s while keeping receive-side power low. The window runs independently of the heartbeat cadence, so critical alarms are detected and transmitted even during the 12 h stopped gap.

Parameter	Value	Note
RX listen window	100 ms	Receiver active to capture a critical-alarm advertisement.
RX interval	5 s	Period between listen windows (sleep in between for battery saving).
Worst-case detection latency	~5 s	A node must advertise repeatedly so it is caught within a window.
Independence	Always on	Runs regardless of running/stopped heartbeat cadence.

On detection the gateway wakes, brings up the link, and publishes the alarm immediately (QoS 1). Routine (non-critical) events and telemetry are not carried on this window — they ride the scheduled heartbeat.

4. Event Message (mt = "event")

Discrete state changes that are operationally significant but not safety alarms. Published QoS 1 on the event topic. TRAIN_START / TRAIN_STOP are also the cadence pivots that flip the heartbeat interval (§5).

4.1 Payload (d) field dictionary

Key	Type	Description
code	string	Event code — see §4.2.
from	string null	Previous state.
to	string null	New state.
node	string null	Source sensor node serial, if applicable.
trip	string null	Trip ID — pairs a TRAIN_START with its TRAIN_STOP.
info	obj	Optional event-specific extras (e.g. geofence id, band asset).

4.2 Event code enumeration

code	Meaning	Clause
TRAIN_START	Motion begins → cadence switches to running (10 min)	§7.4
TRAIN_STOP	Motion ends → cadence switches to stopped (12 h)	§7.4
DOOR_OPEN / DOOR_CLOSE	Authorised, in-zone door state change	§5.5
HANDBRAKE_APPLIED / _RELEASED	Handbrake change while stationary (normal)	§5.6
GEOFENCE_ENTER / _EXIT	Wagon crosses a defined geo-fence	§7.9
LOAD_CHANGE	Loaded / unloaded at yard	§5.3
BAND_CHANGE	Condition band transition (Green/Yellow/Red)	§7.10
CHARGE_START / CHARGE_STOP	Solar charging begins / ends	—
SRC_SWITCH	Power source switchover LTO ↔ backup	—
CONN_RESTORED	Connectivity restored; buffer flush begins	§7.5
BOOT	Gateway power-up / reset	—

4.3 Sample — train start

```
{
  "mt": "event", "pv": 1, "seq": 20990, "gw": "GW000734", "wgn": "BOXN21456789",
  "ts": 1718995600,
  "loc": { "lat": 18.61120, "lon": 73.92010, "cep": 4 },
  "spd": 6,
  "pwr": { "src": "lto", "soc": 76, "sol": "charging", "bkp": "ok" },
  "d": { "code": "TRAIN_START", "from": "stopped", "to": "running",
        "node": null, "trip": "trip-2025-06-21-07", "info": {} }
}
```

4.4 Sample — train stop

```
{
  "mt":"event", "pv":1, "seq":21210, "gw":"GW000734", "wgn":"BOXN21456789",
  "ts":1719020400,
  "loc":{"lat":19.07640, "lon":72.87720, "cep":4 },
  "spd":0,
  "pwr":{"src":"lto", "soc":69, "sol":"off", "bkp":"ok" },
  "d":{"code":"TRAIN_STOP", "from":"running", "to":"stopped",
    "node":null, "trip":"trip-2025-06-21-07",
    "info":{"run_km":312.4, "run_h":6.9 } }
}
```

4.5 Sample — power source switchover

```
{
  "mt":"event", "pv":1, "seq":21255, "gw":"GW000734", "wgn":"BCNA33120097",
  "ts":1719060000,
  "loc":{"lat":19.07601, "lon":72.87766, "cep":5 },
  "spd":0,
  "pwr":{"src":"bkp", "soc":8, "sol":"off", "bkp":"ok" },
  "d":{"code":"SRC_SWITCH", "from":"lto", "to":"bkp",
    "node":null, "trip":null,
    "info":{"reason":"lto_below_switchover", "lto_v":4120 } }
}
```

5. Heartbeat Message (mt = "hb")

Periodic full snapshot. Cadence: **10 min running / 12 h stopped** (§7.4 stationary throttling). Largest message; published QoS 1 (QoS 0 acceptable if buffered records guarantee eventual delivery). Carries the full power block including both backup states.

5.1 Payload (d) field dictionary

Key	Type	Description
mode	running stopped	Current cadence mode.
mot	obj	{ spd (km/h), hdg (deg) } motion vector.
gnss	obj	GNSS engine status: { sys:[...], fix: none 2d 3d, nsat, hdop }. sys = constellations in the current fix (§2.3); nsat = satellites used; hdop = horizontal dilution of precision.
sens	array	Sensor readings; one object per node (§5.2).
band	obj	Condition bands: { brg:[...], whl:[...] } values G Y R (§7.10).
pwr2	obj	Full power status block (§5.3).
gsm	obj	{ cell, rssi } serving cell ID and RSSI dBm.
buf	obj	{ n, old, new } buffered record count, oldest/newest ts (§7.5).
up	num	Uptime hours since last BOOT.
fw	string	Firmware version.

5.2 Sensor reading object (sens[] element)

Key	Description	typ values
node	Sensor node serial	—
typ	Sensor type	btemp, load, door, hbrake, ttemp, whl
val	Reading (numeric, or 0/1 for door/handbrake)	—
unit	Unit (C, mm, bar, state)	—
t	Sensor's own board temperature (C)	—
bat	Sensor node battery remaining (%). Reported in the node's BLE advertisement and relayed by the gateway; trended for predictive maintenance since these cells are non-replaceable. A NODE_LOW_BATTERY alarm fires when it crosses the configured threshold.	—

5.3 Full power status block (pwr2)

rem for the Li-SOCl₂ backup is coulomb-counted, not voltage-inferred — the cell holds ~3.6 V almost to end-of-life, so voltage is meaningless for SoC until the cliff. This field must come from integrated discharge over the cell's life.

Path	Key	Description
—	src	Active rail (lto bkp trn), from the power-mux status.
lto	v	LTO pack voltage, mV.
lto	soc	LTO state-of-charge %.
lto	chg	Charger state: idle trickle fast tophoff done fault.
sol	vin	Solar input (VBUS) mV.
sol	i	Charge current, mA.
sol	st	off present charging.
bkp	ok	Backup present & healthy (bool).
bkp	v	Backup voltage, mV (~3600).
bkp	rem	Backup remaining capacity, mAh (coulomb-counted).
bkp	use	Backup currently sourcing the rail (bool).
—	aut	Estimated autonomy at current draw, hours.

5.4 Sample — heartbeat, running

```
{
  "mt":"hb", "pv":1, "seq":21000, "gw":"GW000734", "wgn":"BOXN21456789",
  "ts":1718996200,
  "loc":{"lat":18.66010, "lon":73.95220, "cep":4 },
  "spd":58,
  "pwr":{"src":"lto", "soc":74, "sol":"charging", "bkp":"ok" },
  "d":{"
    "mode":"running",
    "mot":{"spd":58, "hdg":271 },
    "gnss":{"sys":["navic","gps"], "fix":"3d", "nsat":11, "hdop":0.8 },
    "sens":[
      { "node":"NODE-BRG-01", "typ":"btemp", "val":47, "unit":"C", "t":46, "bat":94 },
      { "node":"NODE-BRG-02", "typ":"btemp", "val":51, "unit":"C", "t":50, "bat":91 },
      { "node":"NODE-LOAD-01", "typ":"load", "val":62, "unit":"%", "t":44, "bat":88 },
      { "node":"NODE-DOOR-01", "typ":"door", "val":0, "unit":"state", "t":43, "bat":73 },
      { "node":"NODE-HB-01", "typ":"hbrake", "val":0, "unit":"state", "t":43, "bat":96 }
    ],
    "band":{"brg":["G","G","G","Y","G","G","G","G"], "whl":["G","G","G","G"] },
    "pwr2":{"
      "src":"lto",
      "lto":{"v":4810, "soc":74, "chg":"fast" },
      "sol":{"vin":3620, "i":410, "st":"charging" },
      "bkp":{"ok":true, "v":3601, "rem":18450, "use":false },
      "aut":2160
    },
    "gsm":{"cell":"404-22-1A2F", "rssi":-71 },
    "buf":{"n":0, "old":0, "new":0 },
    "up":612, "fw":"1.2.0"
  }
}
```

5.5 Sample — heartbeat, stopped (12 h cadence, on backup)

```

{
  "mt":"hb", "pv":1, "seq":21260, "gw":"GW000734", "wgn":"BCNA33120097",
  "ts":1719100800,
  "loc":{"lat":19.07601, "lon":72.87766, "cep":5 },
  "spd":0,
  "pwr":{"src":"bkp", "soc":6, "sol":"off", "bkp":"ok" },
  "d":{"
    "mode":"stopped",
    "mot":{"spd":0, "hdg":0 },
    "sens":[
      { "node":"NODE-D00R-01","typ":"door", "val":0, "unit":"state","t":31, "bat":72 },
      { "node":"NODE-HB-01", "typ":"hbrake","val":1, "unit":"state","t":31, "bat":95 }
    ],
    "band":{"brg":["G","G","G","G","G","G","G","G"], "whl":["G","G","G","G"] },
    "pwr2":{"
      "src":"bkp",
      "lto":{"v":4020, "soc":6, "chg":"idle" },
      "sol":{"vin":0, "i":0, "st":"off" },
      "bkp":{"ok":true, "v":3598, "rem":18020, "use":true },
      "aut":51000
    },
    "gsm":{"cell":"404-22-0C71", "rssi":-95 },
    "buf":{"n":3, "old":1719057600, "new":1719096000 },
    "up":1290, "fw":"1.2.0"
  }
}

```

6. Downlink Command & Response

Cloud issues a command on the dn/cmd topic (QoS 1, queued by the broker while the gateway sleeps). On its next wake the gateway drains the queue, acts, and publishes a resp uplink correlated by cid.

6.1 Command (mt = "cmd", cloud → gateway)

```
{
  "mt": "cmd",
  "cid": "c-88291",           // correlation id echoed in the response
  "cmd": "set_threshold",     // command name (§6.2)
  "args": { "code": "HOT_BEARING", "thr": 95, "unit": "C" },
  "ts": 1718971000
}
```

6.2 Command catalogue

cmd	args	Purpose	Clause
set_threshold	{code,thr,unit}	Set an alarm threshold	§7.15
get_config	{section}	Read back current config	—
set_interval	{run_s,stop_s}	Set reporting cadence	§7.4
set_gnss	{sys:[...]}	Set preferred GNSS constellation set (to obtain a fix)	§7.17
provision_node	{serial,wgn,typ}	Map sensor node to wagon	§7.18-7.19
pull_data	{since,until}	On-demand raw WSN data	§7.20
get_history	{from_seq}	Request buffered records	§7.5
ota_start	{url,ver,size}	Begin firmware update	—
ota_status	{}	Query OTA progress	—
time_sync	{ts}	Set RTC	§7.17
get_status	{}	Force an immediate status snapshot	—
reboot	{}	Restart gateway	—

6.3 Response (mt = "resp", gateway → cloud)

Key	Type	Description
cid	string	Correlation ID copied from the command.
cmd	string	Command name echoed.
res	ok err	Result.
ec	string null	Error code if res = err (e.g. BAD_ARGS, UNSUPPORTED, BUSY).
pl	obj	Returned payload (applied value, config, record batch, OTA state).

Sample — response to set_threshold (ok)

```
{
  "mt":"resp", "pv":1, "seq":21001, "gw":"GW000734", "wgn":"BOXN21456789",
  "ts":1718996260,
  "loc":{"lat":18.66010, "lon":73.95220, "cep":4 },
  "spd":58,
  "pwr":{"src":"lto", "soc":74, "sol":"charging", "bkp":"ok" },
  "d":{"cid":"c-88291", "cmd":"set_threshold", "res":"ok", "ec":null,
    "pl":{"code":"HOT_BEARING", "thr":95, "unit":"C", "applied":true } }
}
```

Sample — response to get_history (buffered batch)

```
{
  "mt":"resp", "pv":1, "seq":21262, "gw":"GW000734", "wgn":"BCNA33120097",
  "ts":1719100860,
  "loc":{"lat":19.07601, "lon":72.87766, "cep":5 }, "spd":0,
  "pwr":{"src":"bkp", "soc":6, "sol":"off", "bkp":"ok" },
  "d":{"cid":"c-90014", "cmd":"get_history", "res":"ok", "ec":null,
    "pl":{"from_seq":21257, "count":3, "next_seq":21260,
      "records":[ /* original buffered envelopes, unchanged seq+ts */ ] } }
}
```

6.4 Command latency note

Because of Class A queuing, a command waits until the gateway's next wake: **≤ 10 min while running, ≤ 12 h while stopped**. A parked wagon does not wake early to poll for commands — queued commands are delivered on the next scheduled heartbeat. Critical alarms are unaffected: they are detected on the receive window (§3.6) and transmitted immediately.

7. MQTT Topic Structure

Base pattern: `smartwagon/v1/{wgn}/{gw}/{dir}/{kind}`. Keying on wagon number first (§7.19) lets the cloud subscribe per-wagon or fleet-wide via wildcards.

7.1 Topic map

Topic	Dir	QoS	Retain	Payload
<code>smartwagon/v1/{wgn}/{gw}/up/alarm</code>	Up	1	No	alarm message
<code>smartwagon/v1/{wgn}/{gw}/up/event</code>	Up	1	No	event message
<code>smartwagon/v1/{wgn}/{gw}/up/hb</code>	Up	1	No	heartbeat message
<code>smartwagon/v1/{wgn}/{gw}/up/resp</code>	Up	1	No	response message
<code>smartwagon/v1/{wgn}/{gw}/dn/cmd</code>	Down	1	No	command message
<code>smartwagon/v1/{wgn}/{gw}/status</code>	Up	1	Yes	LWT / birth: {st,ts}

7.2 Cloud subscription wildcards

Subscription	Receives
<code>smartwagon/v1/+/+/up/alarm</code>	Every alarm, fleet-wide
<code>smartwagon/v1/{wgn}/+/up/#</code>	All uplink for one wagon
<code>smartwagon/v1/+/+/up/hb</code>	All heartbeats (dashboard / fleet health)
<code>smartwagon/v1/+/+/status</code>	Online/offline state of every gateway

7.3 Session, LWT & birth

- **Persistent session:** connect with `clean_session = false` (MQTT 3.1.1) or `Session-Expiry > 0` (MQTT 5). This is what lets the broker hold queued QoS-1 commands on `dn/cmd` while the gateway sleeps — the mechanism behind Class A.
- **LWT:** `smartwagon/v1/{wgn}/{gw}/status` retained, payload `{"st":"offline","ts":...}`. Broker publishes it if the gateway drops without a clean disconnect.
- **Birth:** on connect the gateway publishes `{"st":"online","ts":...}` (retained) to the same topic, then proceeds with its uplink and command drain.

7.4 Wake-cycle sequence (Class A)

Step	Action
1	Wake (scheduled heartbeat tick, or a critical-alarm event detected on the receive window — §3.6).
2	Modem up; MQTT CONNECT (persistent session); publish birth to status.
3	Publish the triggering uplink (up/alarm or up/hb).
4	Broker delivers any queued commands on dn/cmd.
5	Process each command; publish up/resp with matching cid.

Step	Action
6	Flush buffered records oldest-first if connectivity was restored — store-and-forward replay (§8).
7	Clean DISCONNECT; modem off; back to sleep.

8. Offline Buffering & Store-and-Forward

RDSO §7.5 requires the gateway to hold at least one month of data on board when the cellular link is unavailable and deliver it once connectivity returns. This section defines that store-and-forward mechanism: how a link failure is detected, how records are buffered, and how they are replayed without loss or duplication.

8.1 Failure detection

- Loss of cellular registration (modem reports no network), OR
- MQTT CONNECT fails, OR a QoS-1 PUBLISH receives no broker acknowledgement (PUBACK) within the retry timeout. A record is considered **undelivered** until its PUBACK is received.

8.2 What is buffered

Every message that cannot be confirmed delivered is written to non-volatile storage with its **full original envelope intact** — original ts, seq, loc and payload. Heartbeats, events, responses and alarms are all buffered. Alarms are additionally attempted immediately on each wake; buffering is the fallback, not a replacement.

8.3 Replay on reconnect

On the first successful reconnect the gateway drains the buffer in order:

Step	Action
1	Publish birth { st:"online" } to the status topic (§7.3).
2	Emit a CONN_RESTORED event marking the start of the flush (§4.2).
3	Replay buffered records OLDEST-FIRST, in seq order, on their normal topics (up/alarm, up/event, up/hb), each carrying its ORIGINAL ts and seq.
4	Each record is sent QoS 1; it is removed from the buffer only after its PUBACK. At-least-once delivery is guaranteed.
5	If the link drops mid-flush, replay resumes from the first un-acknowledged record on the next reconnect — no gaps, no manual intervention.
6	New live messages take priority and are interleaved ahead of the backlog; a fresh critical alarm is never held behind the queue.

Idempotent by design: because replays reuse the original seq, the cloud de-duplicates on (gw, seq) and re-orders on (ts, seq). Re-sending a record the cloud already has is harmless. The original ts preserves the true event timeline even though delivery is delayed.

8.4 Push vs. pull

- **Push (default):** the gateway autonomously flushes its backlog on reconnect, as above.
- **Pull (on demand):** the cloud may also request a range with get_history { from_seq } (§6.2); the gateway answers with paced batches and a next_seq continuation token until the range is drained. Used for targeted back-fill or audit (§7.20).

8.5 Visibility & overflow policy

- Every heartbeat carries buf { n, old, new } so the cloud always sees how many records remain and the span they cover (§5.1).
- Storage is sized for the 30-day baseline plus headroom. On overflow the buffer behaves as a ring: **oldest heartbeat (telemetry) records are overwritten first; alarms and events are preserved**. An overflow condition is surfaced in the next heartbeat so the cloud knows telemetry gaps may exist. The policy is configurable.
- If the clock was not yet set when a record was captured, its ts is flagged as estimated so the cloud can treat the timestamp accordingly.

9. Spec Cross-Reference (WD-35-MISC-2024)

Clause	Requirement	Where honoured
§5.3.3	Derailment detection & alert	Alarm DERAIL
§5.4.2	Tank temp $-50\dots+150$ °C, instant alert on breach	Alarm TANK_OVERTEMP
§5.5.1	Door open in unauthorised zone / while moving	Alarm DOOR_UNAUTH; Event DOOR_OPEN/CLOSE
§5.6.1	Handbrake engaged while moving	Alarm HANDBRAKE_MOVING; Event HANDBRAKE_*
§7.4	Throttle ping rate when stationary	Heartbeat mode + 10 min/12 h cadence
§7.5	≥ 1 month onboard buffering on loss of link	Store-and-forward replay (§8); buf block; get_history; CONN_RESTORED
§7.9-7.10	Geo-fenced vibration alerts; Green/Yellow/Red bands	band block; Event BAND_CHANGE; sev
§7.15	Vendor-set, runtime-configurable thresholds	set_threshold command
§7.16	Cellular/email alert escalation	Alarm topic → cloud escalation
§7.17	Time/date/location stamp, CEP50 < 5 m	Envelope ts + loc.cep
§7.18-7.19	Node provisioning by serial ↔ wagon number	provision_node command; wgn key
§7.20	On-demand raw WSN data extraction	pull_data command

End of specification — Rev. 1, pv=1. Living document: contents may change by up to ~30% during development; bump pv and the topic v1 segment on any breaking change.